

GOUTHAM ADITYA PELLURI

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Summary

- I'm a Data Engineer with 3+ years of experience designing and deploying scalable data pipelines and cloud-native architectures across AWS and Azure. Proficient in tools like AWS Glue, Redshift, Azure Data Factory, and Databricks, I specialize in building batch and streaming workflows using PySpark, Kafka, and SQL-based transformations.
- I have worked extensively with both structured and semi-structured data to drive insights for real-time analytics, reporting, and decision-making. Skilled at performance tuning, CI/CD automation, and secure data governance, I thrive in cross-functional teams where I can solve complex data challenges and deliver business-ready solutions. I also hold an AWS Data Engineer certification.

Education

Master's in Computer Science, University Of North Carolina At Charlotte

Relevant Coursework: Information Visualisation, Cloud Computing, Big Data Analytics

Charlotte, NC

Bachelor's in Computer Science, Hindustan Institute Of Technology And Sciences

Relevant Coursework: Data Mining, Data Structures & Algorithms, Artificial Intelligence

Chennai, India

Experience

US Bank

Dec 2023 - Present

Associate Data Engineer

Brookfield, USA

- Engineered scalable data pipelines using AWS Glue, Data Flow, Lambda, and Step Functions, streamlining ingestion from heterogeneous sources and reducing ETL execution time by 40%.
- Architected a secure data lake on Amazon S3 with automated lifecycle and compliance features and bucket encryption, enabling GDPR-compliant storage for regulated financial datasets.
- Developed and managed 20+ automated workflows using Airflow and AWS Batch, reducing failure rates by 30% and improving SLA adherence through real-time error detection and recovery mechanisms.
- Reduced manual workload by 50% through automation of ETL scheduling and validation for 10+ pipelines using Python, Shell scripting, and SQL transformation logic.
- Integrated Kinesis for near real-time stream processing to support event-driven analytics from distributed services
- Partnered with support analysts and product managers to resolve real-time data flow issues and align pipeline output with stakeholder KPIs.
- Defined and implemented unit and integration test design frameworks using PyTest for ETL pipelines, ensuring 95% test coverage
- Integrated Informatica and SSIS to modernize ETL pipelines for legacy finance systems, improving pipeline reliability and reducing integration time by 35%, supporting cross-departmental data alignment.

Caliber Technologies

Dec 2021 - July 2023

Data Engineer

Chennai, India

- Developed and maintained scalable ETL pipelines using ADF and Synapse, transforming 1B+ records from multiple sources into analytical models, accelerating BI dashboard refresh rates by 60%.
- Implemented high-throughput Azure Databricks workloads for data transformation and aggregation from Azure Blob Storage and SQL Server, reducing batch processing time by 45%.
- Integrated Event Grid, Azure Functions, and Data Lake Gen2 to streamline ingestion of semi-structured JSON and Parquet data from external APIs and internal apps.
- Secured data assets with RBAC, managed identities, and Private Endpoints, and implemented data masking policies to meet GDPR and SOC 2 compliance.
- Automated infrastructure provisioning and deployment pipelines using Terraform and Azure DevOps, improving deployment speed and environment consistency by 50%.
- Outlined a cloud-based architecture using SAP HANA and SAC to enable real-time dashboards for ops and sales, reducing reporting latency from hours to under 5 minutes for 15+ KPIs.
- Executed end-to-end integrations across SAP, Oracle, AWS, and Redshift, consolidating structured & unstructured datasets (10TB+) and standardizing reporting with star/snowflake models.

Technical Skills

Programming Languages: Python, Java, C++, SQL, JavaScript, HTML, CSS, Typescript
Cloud Platforms:
AWS: S3, Glue, Lambda, Redshift, Kinesis, EMR, CloudWatch
AZURE: Data Factory, Synapse, Databricks, Data Lake Gen2, Event Grid, Azure SQL, Blob Storage, DevOps
Big Data & Distributed Systems: Apache Spark, Hive, Hadoop, Kafka, PySpark, MapReduce, Airflow
Machine Learning & Data Science: Scikit-learn, BeautifulSoup, Statistics, ML Models, Web Scraping
Databases: DynamoDB, PostgreSQL, MySQL, MongoDB, Cassandra, SQL Server, Oracle
Infrastructure & DevOps: Terraform, Git, Jenkins, Kubernetes, Docker, Jira.
BI & Data Modeling: Power BI, Tableau, Star Schema, Snowflake Schema, Dimensional Modeling, Data Lake Design
Libraries & Frameworks: Pandas, NumPy, SQLAlchemy, PyYAML, Boto3, REST APIs, Bootstrap, D3.js
ETL/ELT Tools: Informatica, AWS Glue, Azure Data Factory, SSIS, DataStage, Hevo Data, Matillion, dbt

Projects

- Cloud-Based Sentiment and Trends on Historical Texts** | *Hadoop, Java, Hive, MapReduce, NLP*
- Designed and deployed scalable pipelines using AWS Glue, Lambda, and Step Functions, supporting ingestion from 15+ diverse sources and improving ETL throughput by 40%.
 - Authored custom Hive UDFs and incorporated NLP models for sentiment scoring and bigram extraction, offering decade-wise emotional insights.
 - Refined MapReduce execution through advanced job tuning, boosting processing speed by 35% and minimizing resource overhead.
- AI-Powered Resume Analyzer and Job Matcher using LLMs** | *Python, Hugging Face, FAISS, Streamlit, OpenAI*
- Created an intelligent resume-job matcher utilizing LLM embeddings and GPT-powered feedback, improving job alignment recommendations with 95% accuracy.
 - Integrated FAISS for rapid semantic search and LangChain for modular prompt orchestration, enabling sub-second AI query handling.
 - Built a responsive user interface using Streamlit that enabled 50+ users to upload resumes and receive real-time AI-driven feedback on job-fit and keyword alignment.
- YouTube Data Visualization Using Tableau** | *Tableau, Tableau Prep, Excel, Calculated Fields, Filters, Parameters.*
- Designed a comprehensive Tableau dashboard to visualize YouTube channel performance metrics such as views, likes, comments, subscriber growth, and watch time across multiple time periods.
 - Connected and prepared the dataset using Excel and Tableau Prep, performed data cleaning, calculated engagement ratios (likes-to-views, comments-per-video), and handled outliers to ensure accurate trend analysis.
 - Implemented dynamic filters, parameters, and calculated fields to enable interactive exploration by video category, region, and publish date, helping users identify top-performing content segments.
 - Delivered actionable insights for content strategy, such as optimal posting time, video format effectiveness, and audience engagement trends—empowering creators to refine their publishing approach.
- Fraud Detection System using Spark** | *Apache Spark, PySpark, Kafka, MLlib, Logistic Regression, GBT*
- Led a team of three members to develop a fraud detection system using Spark, processing 50 GB of transaction data.
 - Used Spark’s distributed computing to preprocess data, addressing missing values and outliers, resulting in a 25% data quality enhancement.
 - Achieved exceptional results with machine learning models (Logistic Regression, Random Forest, Gradient-Boosted Trees) showing AUC values 0.99.
 - Implemented real-time fraud detection capabilities with Apache Spark’s Structured Streaming API and Kafka.

Publications & Certifications

AWS Data Engineer - Associate (View Certificate)	Dec 2024
COVID-19-Preventions-Control-System (View Journal)	Nov 2021